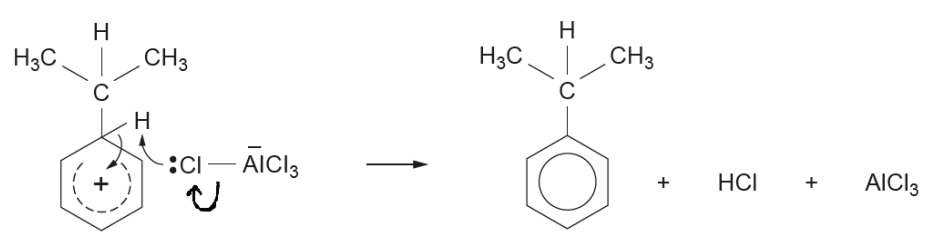


Question				Marking details	Marks available										
					AO1	AO2	AO3	Total	Maths	Prac					
9	(a)	(i)		(concentrated) nitric acid and (concentrated) sulfuric acid	1			1							
		(ii)		tin/iron and (concentrated) hydrochloric acid	1			1							
		(iii)		separation problems – the boiling temperature of the three isomers are too close together accept other sensible answers			1	1							
		(iv)		<table><tr><td>Reagent</td><td>FeCl₃</td><td>NaHCO₃</td></tr><tr><td>Observation</td><td>purple colour</td><td>no change</td></tr></table>	Reagent	FeCl ₃	NaHCO ₃	Observation	purple colour	no change	1	1		2	
Reagent	FeCl ₃	NaHCO ₃													
Observation	purple colour	no change													
	(b)			→ 5C + 2CO + N ₂ + 3H ₂ O		1		1							
	(c)			 award (1) for curly arrows – must have arrow into benzene ring and one other award (1) for all three products	1	1		2							